

Package ‘searchlight’

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Title Audited Analysis of Police Stop and Search Records

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Description Acquire and audit public police stop and search archives for England and Wales, retaining provenance and explicit coverage information. Designed for exposure-based ethnic disparity analysis with separately reported sampling and assumption uncertainty. Contains public sector information licensed under the Open Government Licence v3.0.

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sl_archive_download	<i>Download and verify the necessary archive snapshots</i>
---------------------	--

Description

Selects the newest available snapshots that cover the requested months. Forces are recorded as extraction filters; the source only offers full ZIPs. Checks publisher MD5 before recording local SHA-256. Existing files must match both the publisher and any recorded SHA-256; corrupt files are never replaced silently. The manifest and ZIPs are written to dir.

Usage

```
sl_archive_download(months, forces = NULL, dir = sl_cache_dir(), index = NULL)
```

Arguments

months	Character vector of YYYY-MM months.
forces	Force identifiers, or NULL for all forces.
dir	Cache directory.
index	Optional previously retrieved archive index.

Value

Manifest tibble, or NULL when a resource is unavailable.

See Also

[sl_archive_snapshot\(\)](#), [sl_list_versions\(\)](#)

Other ingest: [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
index <- readRDS(system.file("extdata", "archive-index.rds",
  package = "searchlight"
))
index$archive_file[1]
```

sl_archive_index

List available bulk archive snapshots

Description

Parses advertised month ranges and publisher MD5 checksums. The complete rolling snapshot can contain several years. Results are cached only after this function is called. Source failures return NULL with a message.

Usage

```
sl_archive_index(
  dir = sl_cache_dir(),
  refresh = FALSE,
  url = "https://data.police.uk/data/archive/"
)
```

Arguments

dir	Cache directory.
refresh	Refresh the cached index.
url	Archive listing URL, normally the official source.

Value

A tibble with archive name, snapshot month, coverage range, URL and MD5, or NULL if unavailable.

See Also

[sl_archive_download\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
index <- readRDS(system.file("extdata", "archive-index.rds",
  package = "searchlight"
))
index[, c("archive_file", "month_start", "month_end")]
```

sl_archive_snapshot	<i>Extract immutable stop and search CSV snapshots</i>
---------------------	--

Description

Other archive contents are not extracted. Retains source ZIPs and raw CSV bytes. A manifest records archive and CSV SHA-256, row count and original member path. ZIP traversal paths are refused. Previously extracted CSVs are hash-checked.

Usage

```
sl_archive_snapshot(dir = sl_cache_dir(), months = NULL, forces = NULL)
```

Arguments

dir	Cache containing archive ZIPs and optionally a download manifest.
months	Optional YYYY-MM filter; defaults to the download request.
forces	Optional force filter; defaults to the download request.

Value

A tibble of CSV versions. Writes snapshots and their manifest in dir.

See Also

[sl_select_version\(\)](#), [sl_read_records\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
sl_list_versions(system.file("extdata", "sample", package = "searchlight"))
```

sl_assign_geography	<i>Assign anonymised points to local geographies</i>
---------------------	--

Description

Uses British National Grid (EPSG:27700). Missing or invalid coordinates are retained with NA assignments. On shared polygon edges, ties are resolved by code and flagged ambiguous. Boundary distances describe anonymised snap points, not the unknown true locations; resulting error may be systematic. The boundary-sensitive share uses assigned events with measured distances. Missing-coordinate and unassigned shares use all supplied events.

Usage

```
sl_assign_geography(
  records,
  boundaries,
  types = names(boundaries),
  threshold = 50
)
```

Arguments

records	Contract-bearing records.
boundaries	A named list of sf layers, or one sf layer with metadata.
types	Names of the layers to assign; defaults to supplied layers.
threshold	Boundary-sensitive distance in metres.

Value

Records with geography codes, per-layer distances and flags, and an updated contract. No event rows are dropped or duplicated.

See Also

[sl_boundaries\(\)](#), [sl_location_quality\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
b <- readRDS(system.file("extdata", "sample-boundaries.rds",
  package = "searchlight"
))
x <- sl_sample()[1:3, ]
b$msoa21 <- b$msoa21[b$msoa21$geography_code %in% x$msoa21, ]
sl_assign_geography(x, b, "msoa21")
```

sl_benchmark

Cross-check annual totals against the Home Office

Description

Years end on 31 March. Ratios are withheld unless twelve force-months are submitted or refreshed and the current records contain every selected row. The benchmark is not an equality guarantee: scope of powers, event versus person/vehicle reporting, publication cut-offs and revisions can differ.

Usage

```
sl_benchmark(records, ppap_table = NULL, tolerance = 0.1)
```

Arguments

records	Contract-bearing records.
ppap_table	Table with force_id, year_end and published_total. Default is the bundled Home Office table SS_20, year ending March 2025.
tolerance	Flag absolute proportional differences exceeding this value.

Value

A tibble with observed/published counts, coverage, ratio and flags.

See Also

[sl_coverage\(\)](#)

Other audit: [sl_coverage\(\)](#), [sl_coverage_compare\(\)](#), [sl_location_quality\(\)](#), [sl_quality\(\)](#), [sl_timestamp_quality\(\)](#)

Examples

```
ppap <- utils::read.csv(system.file("extdata", "ppap-totals.csv",
  package = "searchlight"
))
sl_benchmark(sl_sample(), ppap)
```

sl_boundaries	<i>Fetch a versioned ONS boundary layer</i>
---------------	---

Description

Uses a verified catalogue; latest means the latest verified vintage in the installed catalogue, not an unrecorded live change. Boundaries are BGC (20 m generalised, coast clipped). Region polygons cover England. Census hierarchy and current administrative lookups are attached with separate vintages.

Usage

```
sl_boundaries(
  type,
  vintage = "latest",
  dir = sl_cache_dir(),
  codes = NULL,
  lookups = TRUE
)
```

Arguments

type	One of lsoa21, msoa21, ward, lad, pfa, region.
vintage	A catalogue vintage (YYYY-MM), or latest.
dir	Explicit cache directory.
codes	Optional ONS codes to limit acquisition.
lookups	Fetch and attach the ONS lookup tables.

Value

An sf object with geography_code, geography_name, source metadata and lookups, or NULL with a message when the source is unavailable.

See Also

[sl_assign_geography\(\)](#), [sl_population\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
boundaries <- readRDS(system.file("extdata", "sample-boundaries.rds",
  package = "searchlight"
))
boundaries$lsoa21[1, "geography_code"]
```

sl_cache_clear	<i>Clear a marked searchlight cache</i>
----------------	---

Description

Only removes searchlight-managed entries in a directory bearing the marker created by acquisition. Other user files are retained. This is destructive.

Usage

```
sl_cache_clear(dir = sl_cache_dir())
```

Arguments

dir Cache directory.

Value

Invisibly, paths removed.

See Also

[sl_cache_dir\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
d <- tempfile()
dir.create(d)
file.create(file.path(d, ".searchlight-cache"))
sl_cache_clear(d)
unlink(d, recursive = TRUE)
```

sl_cache_dir	<i>Locate the searchlight cache</i>
--------------	-------------------------------------

Description

Returns the cache path without creating it. Acquisition functions explicitly create and write to this directory. Set `options(searchlight.cache_dir = ...)` to override the platform default, or pass an explicit `dir`.

Usage

```
sl_cache_dir()
```

Value

A character path.

See Also

[sl_archive_index\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
sl_cache_dir()
```

sl_contract	<i>Inspect an ingestion contract</i>
-------------	--------------------------------------

Description

Contracts retain source provenance through subsetting and dplyr verbs. The coverage and timestamp tables describe the ingested sources; scope records the currently visible row count and force-months after filtering. Filtering does not turn a submitted file into a missing submission.

Usage

```
sl_contract(x)
```

Arguments

x Records, aggregates or a contract.

Value

A named list of class `sl_contract`. `summary()` returns coverage.

See Also

[sl_read_records\(\)](#), [sl_sample\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
sl_contract(sl_sample())
```

sl_counts

*Count events while preserving submission strata***Description**

Force and month always remain in the result, even if omitted from by: these identify independent reporting and time-exposure cells. Unknown ethnicity remains a level. Missing submissions have NA counts; submitted combinations without events have zero. Filtering records does not narrow the source time grid: pass months and forces explicitly to select the analysis period.

Usage

```
sl_counts(
  records,
  by = c("pfa", "month", "ethnicity_5"),
  units = NULL,
  months = NULL,
  forces = NULL
)
```

Arguments

records	Contract-bearing records.
by	Additional grouping columns, including exactly one ethnicity field.
units	Optional force_id and geography mapping defining the full area universe, including areas without events. Otherwise uses contract units or observed areas. PFA defaults to the published force-code mapping.
months, forces	Explicit analysis scope; defaults to contract coverage.

Value

An sl_counts tibble with canonical geography_code, ethnicity, n, force_id and month columns, original grouping fields, and coverage status.

See Also

[sl_rates\(\)](#), [sl_coverage\(\)](#)

Other rates: [sl_rate_ratio\(\)](#), [sl_rates\(\)](#)

Examples

```
counts <- sl_counts(sl_sample())
head(counts)
```

sl_count_model

*Fit count models with population-time exposure***Description**

Models recorded events, never unique people or a population-minus-stops cell. A population offset is multiplied by months_submitted/12 when available. Other offset columns must already represent exposure over the observed period. Unknown ethnicity and rows without positive exposure are reported as exclusions. Moran's I uses mean Pearson residuals per area and is exploratory after fitting. The exponentiated intercept is a baseline rate per exposure unit; other exponentiated coefficients are multiplicative rate effects. Confidence bounds are reported on this exponentiated scale.

Usage

```
sl_count_model(
  counts,
  formula,
  family = c("poisson", "negbin"),
  offset = "population",
  random = NULL,
  boundaries = NULL,
  conf_level = 0.95
)
```

Arguments

counts	Counts with a contract and exposure column, usually sl_rates.
formula	A formula with response n and desired fixed predictors.
family	Poisson or negative binomial.
offset	Name of the population or population-time exposure column.
random	Optional one-sided lme4 formula, for example ~ (1 force_id).
boundaries	Optional sf polygons keyed by geography_code for residuals.
conf_level	Wald confidence level.

Value

A tidy sl_count_model coefficient table with model, diagnostics and excluded row/event counts in attributes, carrying the ingestion contract.

See Also

[sl_rates\(\)](#), [sl_spatial_disparity\(\)](#)

Other inference: [sl_hit_rates\(\)](#), [sl_simulate\(\)](#), [sl_spatial_disparity\(\)](#), [sl_spatial_map\(\)](#), [sl_veil_of_darkness\(\)](#)

Examples

```
r <- readRDS(system.file("extdata", "example-rates.rds",
  package = "searchlight"
))
r <- r[r$ethnicity %in% c("White", "Black"), ]
sl_count_model(r, n ~ ethnicity)
# Supply matching boundaries to request the residual Moran diagnostic.
```

sl_coverage

Audit submission coverage on a full force-month grid

Description

Counts come from the immutable selected files, including after filtering records. A submitted zero-row CSV is zero; an absent CSV is NA. Refreshes describe differing version counts. Partial suspicion takes precedence over refreshes, and uses preceding calendar months (at most twelve), not twelve observed files. A changelog issue does not create a nonexistent submission.

Usage

```
sl_coverage(
  records,
  forces = NULL,
  months = NULL,
  changelog = NULL,
  partial_threshold = 0.2
)
```

Arguments

records	Contract-bearing records.
forces	Force IDs; default is the contract's requested forces.
months	Months; default is the contract's requested months.
changelog	Table with force_id, month, note and optional issue flag. Default uses the bundled, dated changelog extract.
partial_threshold	Fraction of trailing median, default 0.2.

Value

An sl_coverage tibble with contract; plot gives a coverage heatmap.

See Also

[sl_quality\(\)](#), [sl_coverage_compare\(\)](#), [sl_benchmark\(\)](#)

Other audit: [sl_benchmark\(\)](#), [sl_coverage_compare\(\)](#), [sl_location_quality\(\)](#), [sl_quality\(\)](#), [sl_timestamp_quality\(\)](#)

Examples

```
audit <- sl_coverage(sl_sample())
audit
# plot(audit) displays the submission heatmap.
```

sl_coverage_compare	<i>Compare coverage patterns across two periods</i>
---------------------	---

Description

Periods are explicit month vectors sorted chronologically. Comparability means equal length, matching submitted positions, at least one submitted position, and no partial suspicion. Identical missing positions are listed but allow comparison of the same observed subset. This tests coverage only, not seasonality, reporting practice or population comparability.

Usage

```
sl_coverage_compare(coverage, period_a, period_b)
```

Arguments

coverage	Output of <code>sl_coverage</code> .
period_a, period_b	Month vectors for comparison.

Value

A tibble with comparable, submitted positions, and a list of gaps.

See Also

[sl_coverage\(\)](#)

Other audit: [sl_benchmark\(\)](#), [sl_coverage\(\)](#), [sl_location_quality\(\)](#), [sl_quality\(\)](#), [sl_timestamp_quality\(\)](#)

Examples

```
sl_coverage_compare(sl_coverage(sl_sample()), "2026-05", "2026-06")
```

`sl_denominator_scenarios`*Compare compatible population denominator scenarios*

Description

Exposure tables must have identical geography/classification and cell keys. Each sampling interval is conditional on its exposure scenario; the range of point estimates across scenarios is reported separately.

Usage

```
sl_denominator_scenarios(  
  counts,  
  exposures,  
  reference = "White",  
  comparison = "Black"  
)
```

Arguments

<code>counts</code>	Contract-bearing event counts.
<code>exposures</code>	Named list of <code>sl_exposure</code> tables.
<code>reference, comparison</code>	Groups to compare.

Value

An `sl_sensitivity` tibble with scenario ratios and assumption ranges.

See Also

[sl_exposure\(\)](#), [sl_missing_ethnicity_bounds\(\)](#)

Other sensitivity: [sl_missing_ethnicity_bounds\(\)](#), [sl_ranking_stability\(\)](#), [sl_standardise\(\)](#)

Examples

```
p <- readRDS(system.file("extdata", "sample-population.rds",  
  package = "searchlight"  
)$msoa21  
c <- readRDS(system.file("extdata", "example-counts.rds",  
  package = "searchlight"  
)  
head(sl_denominator_scenarios(c, list(resident = p)))
```

sl_exposure

*Construct an explicit exposure table***Description**

Population is an exposure. Unknown ethnicity has no Census population denominator and must have NA population. Additional age_band and sex columns identify disjoint strata. Counts may exceed exposure.

Usage

```
sl_exposure(
  data,
  geography,
  classification = c("5", "19"),
  source = "user_supplied"
)
```

Arguments

data	A data frame with geography_code, ethnicity and population.
geography	Geography type and vintage, for example msoa21.
classification	Self-defined classification, 5 or 19.
source	Description of a user-supplied exposure.

Value

A tibble with class sl_exposure and an exposure contract.

See Also

[sl_population\(\)](#), [sl_population_crosstab\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
sl_exposure(data.frame(
  geography_code = "area1", ethnicity = "White",
  population = 1000
), geography = "example")
```

sl_hit_rates

Describe three separate outcomes conditional on being searched

Description

Computes each requested outcome separately with Wilson binomial intervals. Unknown ethnicity remains in descriptive tables and missing outcomes are not failures. Logistic comparisons use known ethnicity, force and object fixed effects; constant controls are explicitly omitted. These are associations among searches. Differing hit rates alone do not establish discrimination; selection, differing risk distributions and infra-marginality matter.

Usage

```
sl_hit_rates(
  records,
  by = c("pfa", "object_group"),
  outcome = c("any_action", "arrest", "outcome_linked_to_object"),
  reference = "White",
  comparison = "Black",
  conf_level = 0.95
)
```

Arguments

records	Search records with the three derived binary outcome measures.
by	Descriptive grouping columns; ethnicity_5 is always retained.
outcome	Separate outcomes to calculate (all three by default).
reference, comparison	Known self-defined ethnicity groups.
conf_level	Wilson and logistic Wald confidence level.

Value

An sl_hit_rates tibble. Models, coefficients and exclusions describe separate logistic models. The ingestion contract is retained.

See Also

[sl_read_records\(\)](#), [sl_veil_of_darkness\(\)](#)

Other inference: [sl_count_model\(\)](#), [sl_simulate\(\)](#), [sl_spatial_disparity\(\)](#), [sl_spatial_map\(\)](#), [sl_veil_of_darkness\(\)](#)

Examples

```
hits <- sl_hit_rates(sl_sample())
head(hits)
attr(hits, "coefficients")
```

sl_list_versions	<i>List force-month archive versions</i>
------------------	--

Description

List force-month archive versions

Usage

```
sl_list_versions(dir = sl_cache_dir())
```

Arguments

dir A snapshot cache or bundled sample directory.

Value

A tibble with each force-month's archive, CSV hash and row count.

See Also

[sl_archive_snapshot\(\)](#), [sl_select_version\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
sl_list_versions(system.file("extdata", "sample", package = "searchlight"))
```

sl_location_quality	<i>Diagnose anonymised location quality</i>
---------------------	---

Description

Snap-point concentration is the share at the most common reported coordinate within an LSOA, among records with usable coordinates in that LSOA. It is not the concentration at true incident locations. Missing LSOA assignments remain a separate group. Per-geography boundary diagnostics remain in the contract.

Usage

```
sl_location_quality(records)
```

Arguments

records Contract-bearing records, optionally with assigned LSOAs.

Value

A force-month-LSOA tibble with missingness, concentration and boundary sensitivity, carrying the contract.

See Also

[sl_assign_geography\(\)](#), [sl_timestamp_quality\(\)](#)

Other audit: [sl_benchmark\(\)](#), [sl_coverage\(\)](#), [sl_coverage_compare\(\)](#), [sl_quality\(\)](#), [sl_timestamp_quality\(\)](#)

Examples

```
sl_location_quality(sl_sample()[1:100, ])
```

sl_missing_ethnicity_bounds

Bound disparity under allocations of Unknown ethnicity

Description

Extreme allocations send every Unknown event to one of the two groups. Proportional allocation uses all known groups, not only the comparison pair. `force_object_mar` uses known-group shares within force and object, pooled over areas and months; it refuses strata with no known ethnicity. These are assumptions, not confidence intervals. Force/object allocation also assumes pooled known composition applies to each recipient area/month; MAR alone does not establish that transportability. The tipping share solves the equation where the ratio equals one, with all remaining Unknown sent to the reference. A tipping value outside zero to one means no feasible tipping allocation.

Usage

```
sl_missing_ethnicity_bounds(
  counts,
  population,
  reference = "White",
  comparison = "Black",
  scenarios = c("all_to_reference", "all_to_comparison", "proportional",
    "force_object_mar")
)
```

Arguments

<code>counts</code>	Contract-bearing event counts.
<code>population</code>	Marginal ethnic-group exposure table.
<code>reference, comparison</code>	Groups to compare.
<code>scenarios</code>	Allocation assumptions to report.

Value

An sl_sensitivity tibble with scenario ratios, extreme bounds, tipping allocation and separately labelled baseline sampling intervals.

See Also

[sl_rates\(\)](#), [sl_denominator_scenarios\(\)](#)

Other sensitivity: [sl_denominator_scenarios\(\)](#), [sl_ranking_stability\(\)](#), [sl_standardise\(\)](#)

Examples

```
c <- readRDS(system.file("extdata", "example-counts.rds",
  package = "searchlight"
))
p <- readRDS(system.file("extdata", "sample-population.rds",
  package = "searchlight"
))$msoa21
head(sl_missing_ethnicity_bounds(c, p))
```

sl_population

Census 2021 ethnicity populations

Description

Fetches TS021 (NM_2041_1), with exact codes verified against returned cells. Supply ONS codes or an sf boundary object. Current ward and LAD boundaries must not be silently matched to Census 2021 units: unchanged codes alone do not demonstrate unchanged boundaries. A new exposure source must be supplied for changed units. Census disclosure control may cause small inconsistencies between separately published tables. Unknown is retained with NA exposure.

Usage

```
sl_population(geography, classification = c("5", "19"), dir = sl_cache_dir())
```

Arguments

geography ONS geography codes, or an sf layer with geography_code.
classification Census classification, 5 or 19.
dir Explicit cache directory, written only on this call.

Value

An sl_exposure tibble, or NULL with a message if unavailable.

See Also

`sl_exposure()`, `sl_population_crosstab()`

Other ingest: `sl_archive_download()`, `sl_archive_index()`, `sl_archive_snapshot()`, `sl_assign_geography()`, `sl_boundaries()`, `sl_cache_clear()`, `sl_cache_dir()`, `sl_contract()`, `sl_exposure()`, `sl_list_versions()`, `sl_population_crosstab()`, `sl_read_records()`, `sl_sample()`, `sl_select_version()`

Examples

```
p <- readRDS(system.file("extdata", "sample-population.rds",
  package = "searchlight"
))
head(p$msoa21)
```

`sl_population_crosstab`

Census ethnicity by age and sex

Description

Uses RM032 (NM_2132_1), available at Census 2021 small areas. Five published age bands are combined to under 25, 25-34 and 35+, the common partition with police age bands. Census sex and police-recorded gender are different measurements; any analysis using them must state this assumption. Unsupported geography returns NULL and a message. No age-sex cross-tabulation is imputed from marginal totals.

Usage

```
sl_population_crosstab(
  geography,
  dir = sl_cache_dir(),
  classification = c("5", "19")
)
```

Arguments

`geography` ONS geography codes, or an sf layer with `geography_code`.
`dir` Explicit cache directory, written only on this call.
`classification` Census classification, 5 or 19.

Value

An `sl_exposure` tibble with `age_band` and `sex`, or NULL.

See Also

[sl_population\(\)](#), [sl_exposure\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
p <- readRDS(system.file("extdata", "sample-crosstab.rds",
  package = "searchlight"
))
head(p)
```

sl_quality

Attach submission, timestamp and location diagnostics

Description

Pass population to complete the population metadata in the records contract; population counts remain a separate exposure table. Diagnostics describe the current records; coverage continues to describe the original source files.

Usage

```
sl_quality(records, population = NULL, threshold = 0.05, changelog = NULL)
```

Arguments

records	Contract-bearing records.
population	Optional sl_exposure table.
threshold	Midnight-share threshold.
changelog	Optional changelog table passed to sl_coverage.

Value

Records with an updated contract and attached quality tables.

See Also

[sl_coverage\(\)](#), [sl_timestamp_quality\(\)](#), [sl_location_quality\(\)](#)

Other audit: [sl_benchmark\(\)](#), [sl_coverage\(\)](#), [sl_coverage_compare\(\)](#), [sl_location_quality\(\)](#), [sl_timestamp_quality\(\)](#)

Examples

```
records <- sl_quality(sl_sample()[1:100, ])
sl_contract(records)$timestamps
```

sl_ranking_stability *Assess area rank uncertainty under count models or posterior draws*

Description

Bootstrap draws event counts from the fitted Poisson or negative-binomial model, then ranks event-rate ratios (largest is rank 1). Quasi-Poisson does not define a count distribution and is refused. Positive totals in both groups are required for plug-in bootstrap; sparse zero-count areas need a suitable posterior model. Ties receive random ranks under the saved seed. Optional scenario rows are analysed separately, never pooled into a sampling distribution. Pairwise stability means ordering probability above 0.95.

Usage

```
sl_ranking_stability(
  rate_ratios,
  method = c("bootstrap", "posterior"),
  n = 1000,
  seed = 1
)
```

Arguments

rate_ratios	An sl_rate_ratio table, or a contract-bearing table with geography_code and a posterior_draws matrix attribute (draws by area).
method	Parametric bootstrap or posterior ranks.
n	Number of draws; default 1000.
seed	Reproducible seed, restored on exit.

Value

An sl_sensitivity tibble of median/95% rank intervals, with attributes rank_probabilities (matrix) and pairwise (ordering probabilities).

See Also

[sl_rate_ratio\(\)](#), [sl_denominator_scenarios\(\)](#)

Other sensitivity: [sl_denominator_scenarios\(\)](#), [sl_missing_ethnicity_bounds\(\)](#), [sl_standardise\(\)](#)

Examples

```
# Posterior draws can also be supplied by an independently fitted model.
x <- tibble::tibble(geography_code = c("a", "b"))
attr(x, "contract") <- sl_contract(sl_sample())
attr(x, "posterior_draws") <- cbind(a = c(1, 2, 3), b = c(3, 2, 1))
sl_ranking_stability(x, "posterior", n = 3)
```

sl_rates

Calculate rates using observed population-time exposure

Description

The primary rate is annualised events per 1,000 resident person-years: $n / (\text{population} * \text{submitted_months} / 12) * \text{per}$. The separate `period_rate` is $n / \text{population} * \text{per}$ for that cell. Neither is a probability of a person being searched. NA submission counts are excluded from time exposure. Partial submissions remain flagged and describe reported events only.

Usage

```
sl_rates(counts, population, per = 1000)
```

Arguments

<code>counts</code>	Output from <code>sl_counts</code> .
<code>population</code>	A compatible <code>sl_exposure</code> table.
<code>per</code>	Rate scaling, default 1000.

Value

An `sl_rates` tibble with `population`, `exposure`, `rate` and `period_rate`.

See Also

[sl_counts\(\)](#), [sl_rate_ratio\(\)](#), [sl_exposure\(\)](#)

Other rates: [sl_counts\(\)](#), [sl_rate_ratio\(\)](#)

Examples

```
c <- readRDS(system.file("extdata", "example-counts.rds",  
  package = "searchlight"))  
p <- readRDS(system.file("extdata", "sample-population.rds",  
  package = "searchlight"))$msoa21  
head(sl_rates(c, p))
```

sl_rate_ratio

Estimate an event stop-rate ratio with exposure offsets

Description

Fits $n \sim \text{ethnicity}$ with $\log(\text{population-time})$ offset within each area and other grouping stratum, pooling submitted months. Poisson intervals are exact conditional intervals for two event counts; zero counts are supported without adding pseudocounts. Quasi-Poisson and negative binomial require replicated cells and use model-based log intervals. Dispersion is Pearson chi-square divided by residual degrees of freedom. This is not a person-level risk ratio.

Usage

```
sl_rate_ratio(
  rates,
  reference = "White",
  comparison = "Black",
  method = c("poisson", "quasipoisson", "negbin"),
  conf_level = 0.95
)
```

Arguments

rates	Output from sl_rates.
reference, comparison	Self-defined ethnicity groups.
method	Count model family.
conf_level	Sampling confidence level.

Value

An sl_rate_ratio tibble with sampling intervals, dispersion, counts, exposures and fitted model list column, carrying the ingestion contract.

See Also

[sl_rates\(\)](#), [sl_missing_ethnicity_bounds\(\)](#)
 Other rates: [sl_counts\(\)](#), [sl_rates\(\)](#)

Examples

```
r <- readRDS(system.file("extdata", "example-rates.rds",
  package = "searchlight"
))
head(sl_rate_ratio(r))
```

sl_read_records	<i>Read selected immutable archive records</i>
-----------------	--

Description

Validates the fixed CSV schema, checksum and row count, preserving every event (including identical rows) and raw ethnicity strings. Timestamps retain their supplied UTC offsets and are displayed in Europe/London. Absent or refused ethnicity is Unknown. Missing outcome measures remain NA, not negative outcomes. The filename defines the submission month. Offset conversion can cross a month boundary; date_month_crossing records this without moving the event into another source file. A timestamp must match the filename in either its supplied clock or the London clock.

Usage

```
sl_read_records(dir = sl_cache_dir(), versions = NULL)
```

Arguments

dir	Snapshot cache directory.
versions	One selected version per force-month; defaults to latest.

Value

An sl_records tibble with a validated ingestion contract.

See Also

[sl_contract\(\)](#), [sl_select_version\(\)](#), [sl_sample\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_sample\(\)](#), [sl_select_version\(\)](#)

Examples

```
d <- system.file("extdata", "sample", package = "searchlight")
v <- sl_select_version(sl_list_versions(d))
records <- sl_read_records(d, v[1, ])
summary(sl_contract(records))
```

sl_report

Write an offline report of audited records and analysis results

Description

Renders the installed Markdown/Rmd template to a self-contained HTML file. No Pandoc, network, model fitting or external web assets are required. Supplied estimates must retain the same source snapshots and ethnicity classification as records. Their analysis scope and diagnostics are shown. Coverage describes source files; filtering does not redefine coverage. Sampling intervals and assumption ranges have separate labelled sections. The report includes a summary, section navigation and expandable audit details. Tables retain their original column names in header tooltips.

Usage

```
sl_report(
  records,
  file,
  estimates = list(),
  title = "Stop and search: evidence and assumptions",
  assumptions = character(),
  limitations = character(),
  max_rows = 500,
  overwrite = FALSE
)
```

Arguments

records	Contract-bearing search records.
file	Explicit HTML output path. Its parent directory must exist.
estimates	Uniquely named list of contract-bearing result tables.
title	Report title, treated as text.
assumptions, limitations	Additional statements, treated as text.
max_rows	Maximum displayed rows per table; truncation is labelled.
overwrite	Whether to replace an existing output file.

Value

Invisibly, a one-row tibble with the output path and record count, carrying the ingestion contract. Writes only the requested output file.

See Also

[sl_coverage\(\)](#), [sl_rate_ratio\(\)](#), [sl_missing_ethnicity_bounds\(\)](#)

Examples

```

rates <- readRDS(system.file("extdata", "example-rates.rds",
  package = "searchlight"
))
path <- tempfile(fileext = ".html")
sl_report(sl_sample(), path, estimates = list(ratios = sl_rate_ratio(rates)))
file.exists(path)

```

sl_sample	<i>Load the bundled real archive sample</i>
-----------	---

Description

Load the bundled real archive sample

Usage

```
sl_sample()
```

Value

An sl_records tibble for the documented sample forces and months.

See Also

[sl_read_records\(\)](#), [sl_contract\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_select_version\(\)](#)

Examples

```

x <- sl_sample()
table(x$force_id, x$month)

```

sl_select_version	<i>Select one complete version per force-month</i>
-------------------	--

Description

Never deduplicates event rows. Ties are resolved by snapshot month and archive name. max_rows chooses the newest version among equal row counts. Differences greater than 1% of the smallest count are reported (zero to positive is infinite).

Usage

```
sl_select_version(
  versions,
  rule = c("latest", "earliest", "max_rows", "manual"),
  manual = NULL
)
```

Arguments

versions	Output from sl_list_versions() .
rule	Version selection rule.
manual	For the manual rule, a tibble with force_id, month, archive_file, containing exactly one choice per force-month.

Value

Selected versions with list-column alternatives and a differences attribute. Each row carries the selection rule.

See Also

[sl_read_records\(\)](#)

Other ingest: [sl_archive_download\(\)](#), [sl_archive_index\(\)](#), [sl_archive_snapshot\(\)](#), [sl_assign_geography\(\)](#), [sl_boundaries\(\)](#), [sl_cache_clear\(\)](#), [sl_cache_dir\(\)](#), [sl_contract\(\)](#), [sl_exposure\(\)](#), [sl_list_versions\(\)](#), [sl_population\(\)](#), [sl_population_crosstab\(\)](#), [sl_read_records\(\)](#), [sl_sample\(\)](#)

Examples

```
d <- system.file("extdata", "sample", package = "searchlight")
v <- sl_list_versions(d)
sl_select_version(v)
```

sl_simulate

Simulate event counts with independent disparity and total intensity

Description

The disparity surface varies east-west and expected total events north-south. Their lattice correlation is zero by construction. Counts are Poisson events, then unknown ethnicity is generated by binomial thinning. MCAR has a common probability, MAR varies by synthetic force, and MNAR varies by ethnicity. True latent rate ratios and complete realised event ratios are distinct.

Usage

```
sl_simulate(
  side = 6,
  surface = c("smooth", "discontinuous", "small"),
  population = c(White = 5000, Black = 1000),
  rate = 0.025,
  missingness = c("none", "mcar", "mar", "mnar"),
  missing_rate = 0.2,
  months = 12,
  seed = 1
)
```

Arguments

side	Lattice width and height, at least three.
surface	Smooth, discontinuous, or small comparison-population setting.
population	Resident reference/comparison populations per area.
rate	Mean total annual event rate per resident.
missingness	Missing-ethnicity mechanism.
missing_rate	Baseline missing probability.
months	Number of submitted months, between one and twelve.
seed	Reproducible seed, restored on exit.

Value

A list of counts, `complete_counts`, population, boundaries, truth and settings. Counts carry a clearly synthetic ingestion contract.

See Also

[sl_spatial_disparity\(\)](#), [sl_missing_ethnicity_bounds\(\)](#)

Other inference: [sl_count_model\(\)](#), [sl_hit_rates\(\)](#), [sl_spatial_disparity\(\)](#), [sl_spatial_map\(\)](#), [sl_veil_of_darkness\(\)](#)

Examples

```
s <- sl_simulate(side = 3, missingness = "mnar", missing_rate = 0.3)
s$truth
sl_missing_ethnicity_bounds(s$counts, s$population,
  scenarios = c("all_to_reference", "all_to_comparison")
)
```

sl_spatial_disparity *Estimate a multivariate spatial stop-rate disparity surface*

Description

Uses CARBayes MVS.CARleroux with two Poisson responses and a matrix of log population-time offsets. Each ethnicity has its own spatial effect ϕ_{ig} . The equivalent shared/contrast decomposition is $u_i = (\phi_{ir} + \phi_{ic})/2$, $v_{ir} = (\phi_{ir} - \phi_{ic})/2$ and $v_{ic} = -v_{ir}$. These components have the covariance induced by the multivariate prior; they are not separately independent priors. The full posterior ratio is $\exp(\alpha_c - \alpha_r + \phi_{ic} - \phi_{ir})$. Smoothing stabilises under stated assumptions; it cannot repair exposure, missing ethnicity, missing submissions or anonymised location errors.

Usage

```
sl_spatial_disparity(
  counts,
  population,
  boundaries,
  reference = "White",
  comparison = "Black",
  backend = c("carbayes", "inla"),
  burnin = 2000,
  n.sample = 12000,
  thin = 10,
  chains = 2,
  k = 2,
  adjacency = NULL,
  boundary_threshold = 0.2,
  seed = 1,
  ...
)
```

Arguments

counts	Marginal event counts with a valid contract.
population	Compatible marginal ethnicity exposure table.
boundaries	sf polygons with geography_code.
reference, comparison	Known self-defined ethnicity groups.
backend	CARBayes. INLA is explicitly deferred beyond version 0.1.0.
burnin, n.sample, thin	MCMC controls per chain, including burn-in in n.sample.
chains	Number of independent chains, at least two, run on one core.
k	Additional ratio threshold for exceedance probability.

adjacency	Optional symmetric adjacency matrix named with geography codes. Otherwise rook adjacency is used; islands require an explicit decision.
boundary_threshold	Warn if the assigned boundary-sensitive share exceeds this proportion (default 0.2).
seed	Reproducible seed, restored on exit.
...	Prior and sampler arguments for MVS.CARleroux, such as rho or MALA.

Value

An `sl_spatial` tibble with 90/95 percent intervals, exceedance probabilities, crude ratios, rank-normalised R-hat and bulk/tail ESS. Joint posterior draws, fitted model, diagnostics and boundaries are attributes.

See Also

`sl_simulate()`, `sl_spatial_map()`, `sl_ranking_stability()`

Other inference: `sl_count_model()`, `sl_hit_rates()`, `sl_simulate()`, `sl_spatial_map()`, `sl_veil_of_darkness()`

Examples

```
# A precomputed synthetic fit avoids MCMC in installed examples.
path <- system.file("extdata", "example-spatial.rds",
  package = "searchlight"
)
if (nzchar(path)) {
  fit <- readRDS(path)
  head(fit)
  head(sl_spatial_map(fit))
}
```

<code>sl_spatial_map</code>	<i>Join spatial posterior summaries to their polygons</i>
-----------------------------	---

Description

Join spatial posterior summaries to their polygons

Usage

```
sl_spatial_map(x)
```

Arguments

`x` An `sl_spatial` fit.

Value

An sf object with summaries and its ingestion contract.

See Also

[sl_spatial_disparity\(\)](#)

Other inference: [sl_count_model\(\)](#), [sl_hit_rates\(\)](#), [sl_simulate\(\)](#), [sl_spatial_disparity\(\)](#), [sl_veil_of_darkness\(\)](#)

Examples

```
path <- system.file("extdata", "example-spatial.rds",
  package = "searchlight"
)
if (nzchar(path)) head(sl_spatial_map(readRDS(path)))
```

sl_standardise	<i>Directly standardise age-sex stop-event rates</i>
----------------	--

Description

Uses common age-sex weights across ethnicity groups. Census sex is compared with police-recorded gender only under an explicit measurement assumption. Unknown age, gender and ethnicity are retained in exclusion diagnostics. A positively weighted stratum with zero exposure makes the standardised rate undefined; it is never silently dropped. Sampling intervals combine exact Poisson stratum intervals with a Bonferroni correction and fixed weights. They are conservative, including when every observed stratum count is zero.

Usage

```
sl_standardise(
  counts,
  crosstab,
  standard = c("england_wales", "study_population"),
  per = 1000
)
```

Arguments

counts	Counts grouped by age_band and sex.
crosstab	Compatible Census or explicit ethnicity-age-sex exposure.
standard	England/Wales Census 2021 or pooled study population weights.
per	Rate scaling, default 1000 per person-year.

Value

An sl_sensitivity tibble with crude and standardised rates, sampling intervals, excluded-event counts, weights and the ingestion contract.

See Also

[sl_population_crosstab\(\)](#), [sl_rates\(\)](#)

Other sensitivity: [sl_denominator_scenarios\(\)](#), [sl_missing_ethnicity_bounds\(\)](#), [sl_ranking_stability\(\)](#)

Examples

```
p <- readRDS(system.file("extdata", "sample-crosstab.rds",
  package = "searchlight"
))
c <- readRDS(system.file("extdata", "example-demographic-counts.rds",
  package = "searchlight"
))
head(suppressWarnings(sl_standardise(c, p)))
```

sl_timestamp_quality *Diagnose timestamp completeness and midnight heaping*

Description

Reports both London-clock midnight and midnight in the source string. Some date-only summer records carry a UTC offset, so checking London midnight alone would miss them. Reliability requires both shares below threshold and no missing timestamps. This is a screening diagnostic, not proof that all supplied times are accurate. Empty and absent force-months are not reliable.

Usage

```
sl_timestamp_quality(records, threshold = 0.05)
```

Arguments

records	Contract-bearing records.
threshold	Maximum permitted midnight share, default 0.05.

Value

A tibble with force-month diagnostics and within-force share range.

See Also

[sl_quality\(\)](#), [sl_location_quality\(\)](#)

Other audit: [sl_benchmark\(\)](#), [sl_coverage\(\)](#), [sl_coverage_compare\(\)](#), [sl_location_quality\(\)](#), [sl_quality\(\)](#)

Examples

```
sl_timestamp_quality(sl_sample())
```

sl_veil_of_darkness	<i>Diagnose ethnicity composition in daylight and darkness</i>
---------------------	--

Description

Excludes every force-month that fails or lacks the contract's time-quality screen, retaining an explicit exclusion table. Annual evening clock-time bounds use each distinct location and year. Civil darkness begins at dusk; ambiguous sunset-to-dusk observations are excluded. Times and clock changes use Europe/London, including British summer time. The logistic comparison controls clock time (natural spline), weekday, month and force where these vary. The DST option further restricts to weeks around both clock changes and adds transition and running-day controls; it is a local darkness comparison. The method was developed for vehicle stops. Transfer to pedestrian searches requires assumptions about visibility, activity, deployment and selection.

Usage

```
sl_veil_of_darkness(
  records,
  boundaries = NULL,
  twilight = c("civil", "sunset"),
  window = "intertwilight",
  design = c("intertwilight", "dst"),
  reference = "White",
  comparison = "Black",
  dst_weeks = 3,
  conf_level = 0.95
)
```

Arguments

records	Contract-bearing search records.
boundaries	Optional sf study region; restrict published points to it. Missing locations are never replaced by centroids.
twilight	Civil twilight end or sunset definition of darkness.
window	Currently only intertwilight, the annual evening overlap window.
design	Annual intertwilight or local DST-window comparison.
reference, comparison	Known self-defined ethnicity groups.
dst_weeks	Weeks on either side of each clock change for the DST design.
conf_level	Logistic Wald confidence level.

Value

An `sl_veil` tibble with an odds ratio, interval and fit status. Attributes `steps`, `excluded_force_months`, `assumptions`, `data` and `model` retain the full design audit. A gated or unidentifiable sample has an undefined ratio.

See Also

[sl_timestamp_quality\(\)](#), [sl_hit_rates\(\)](#)

Other inference: [sl_count_model\(\)](#), [sl_hit_rates\(\)](#), [sl_simulate\(\)](#), [sl_spatial_disparity\(\)](#), [sl_spatial_map\(\)](#)

Examples

```
# Date-only records are explicitly refused by the time-quality screen.
x <- sl_sample()[1:10, ]
day <- as.Date(x$date, tz = "Europe/London")
x$date <- as.POSIXct(paste(day, "00:00:00"), tz = "Europe/London")
x$date_raw <- format(x$date, "%Y-%m-%dT%H:%M:%S%z")
attr(x, "contract")$timestamps <- sl_timestamp_quality(x)
sl_veil_of_darkness(x)
```

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